

多模式FFT处理器Innovus后端流程

FloorPlan

PnR

Foundation Flow Wizard

结果

Place

Route

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只是记录了一下很简单的流程，一切都是最简化的，是笔者用来跑通流程的初次尝试，所以可能会有很多错误。仅供学习参考！

工具版本以及系统版本：

RockyLinux8.10

INNOVUS231、INNOVUS201

GCC 8.5.0

PDK: TSMC N28

PnR 的通用流程：FloorPlan -> Place -> CTS -> Route -> Signoff

FloorPlan

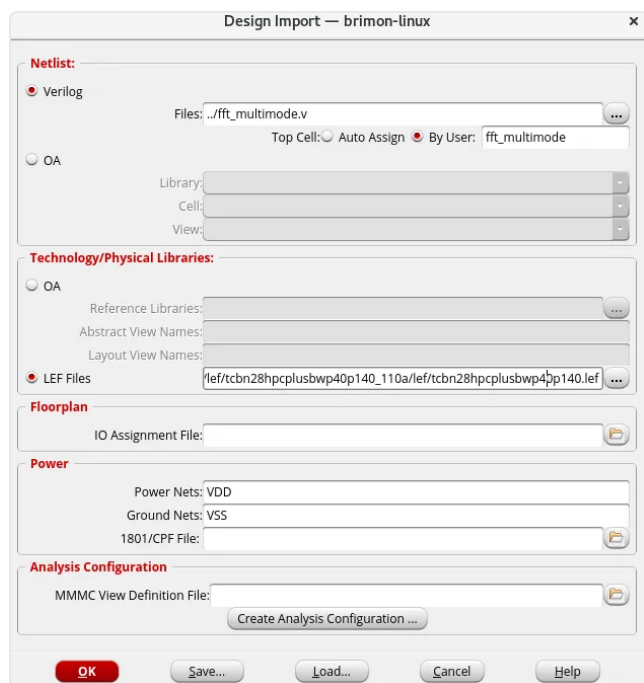
我为了简化后端的流程（先跑通试试），没有综合 Foundry SRAM，所以不用做 macro 的 floorplan 和连线。

第一步要先自己进行 FloorPlan，可以通过 gui 操作或者脚本进行自动 floorplan（由于没有研究明白如何使用脚本进行 fp，所以就手动 fp 了）。

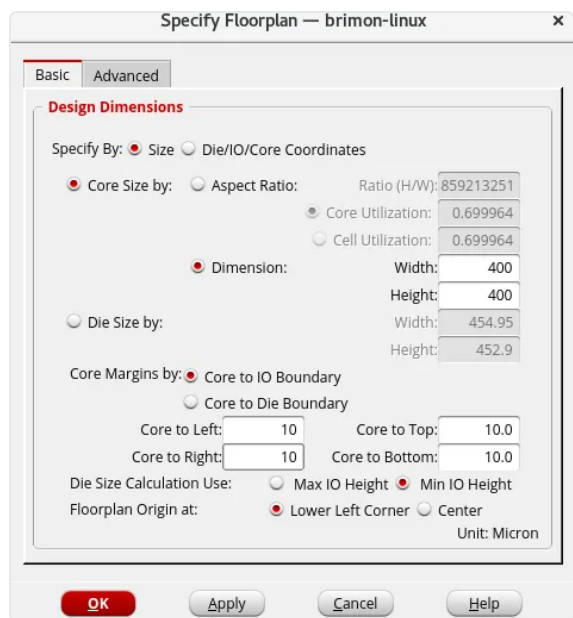
先将综合的结果（.sdc 和.v）拷贝到 APR/innovus 文件夹下

这里用 gui 进行 fp：

先在 innovus 中导入设计，选择 verilog 文件和 lef 文件（tsmc28 一定要导入 telf 和 lef 两个才行），并设置电源名称：

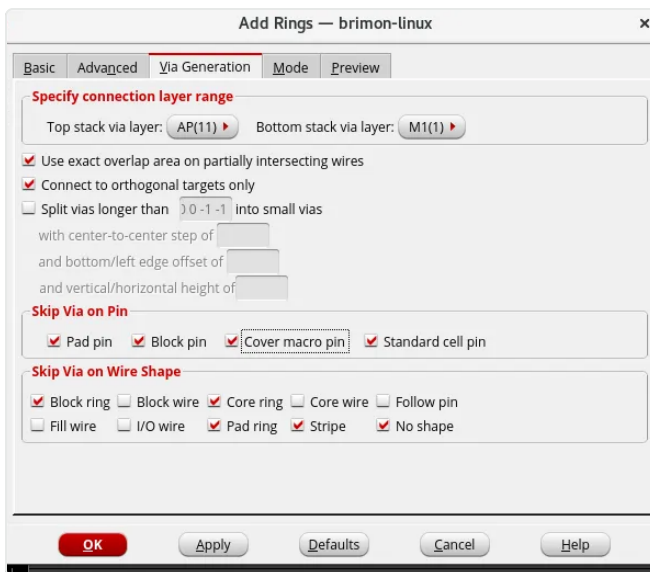
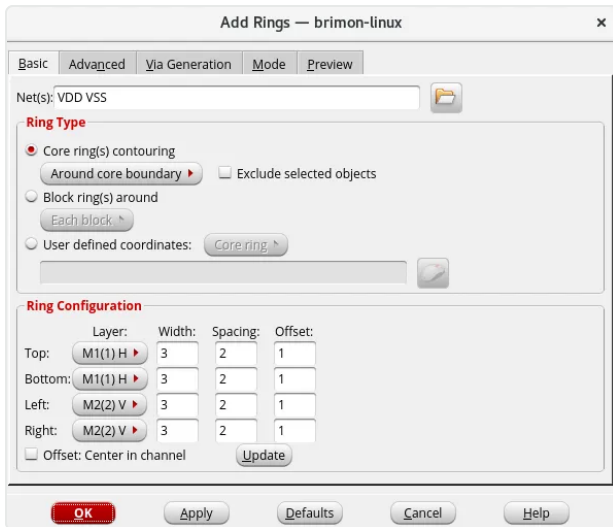


然后创建 floorplan，我这里随便填了几个数，比综合给出的面积要大一些，364x364：



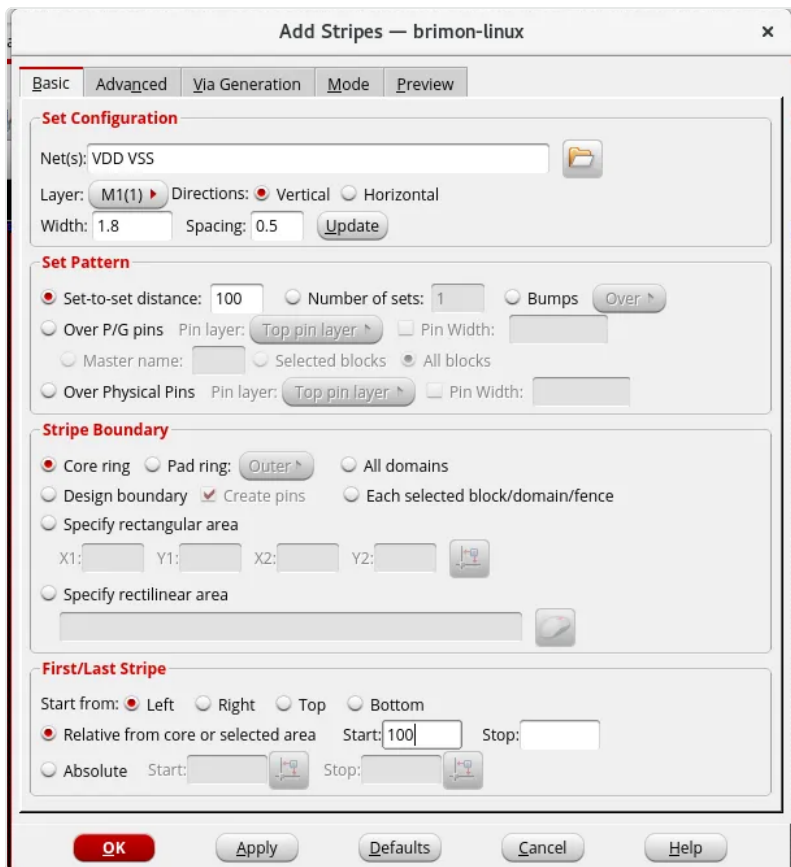
如果用了 Foundry SRAM，这里还需要放置 SRAM，并设置 SRAM 的 Halo。

然后添加 Ring（环绕芯片的电源线）：



在芯片面积或者说规模比较大的情况下，只有Core PG Rings以及Macro PG Rings是不够的，容易产生电压降（IR-drop）问题。我们一般会在Core PG Rings中间的区域加上一些电源地条（PG Strap/Stripe），添加的原则是和Core PG Rings一样尽量在最高层添加，要满足IR-drop的要求，在不影响Routing Sources的情况下加的越多越好。

添加 stripe（竖着的电源线）：



Core区域的所有标准单元都是通过PG Rail来进行供电的，标准单元上下方供电的PG Pin称为Follow Pin。我们需要用特殊绕线（Special Route）来将所有标准单元的Follow Pin都连在一起，形成轨道一样的效果，两侧接到Core PG Ring上。

进行 sroute（横着的电源线）：

执行命令：

```
1 clearGlobalNets
2 globalNetConnect VDD -type pgpin -pin VDD -inst * -module {}
3 globalNetConnect VDD -type tiehi -pin VDD -inst * -module {}
4 globalNetConnect VDD -type net -net VDD -inst * -module {}
5 globalNetConnect VSS -type pgpin -pin VSS -inst * -module {}
6 globalNetConnect VSS -type tielo -pin VSS -inst * -module {}
7 globalNetConnect VSS -type net -net VSS -inst * -module {}
```

然后设置sroute：

SRoute — brimon-linux

Basic Advanced **Via Generation**

Net(s): VDD VSS

SRoute

☒ Block Pins ☒ Pad Rings ☒ Floating Stripes
☒ Pad Pins ☒ Follow Pins ☐ Secondary Power Pins

Routing Control

Layer Change Control

Top Layer: AP(11) Bottom Layer: M1(1)
☒ Allow Jogging ☒ Allow Layer Change

☐ Specify Area

X1: Y1: 
X2: Y2: 

☐ Connect to Target Inside The Area Only

Power Domain Selection

☒ All ☐ Selected
☐ Named:

☐ Delete Existing Routes

Mode Setup Target Editing Options

OK Apply Defaults Cancel Help

SRoute — brimon-linux

Basic Advanced **Via Generation**

Specify Layer Ranges

Crossover Connection:
Top Stack Via Layer: AP(11) Bottom Stack Via Layer: M1(1)
Target Connection:
Top Stack Via Layer: AP(11) Bottom Stack Via Layer: M1(1)

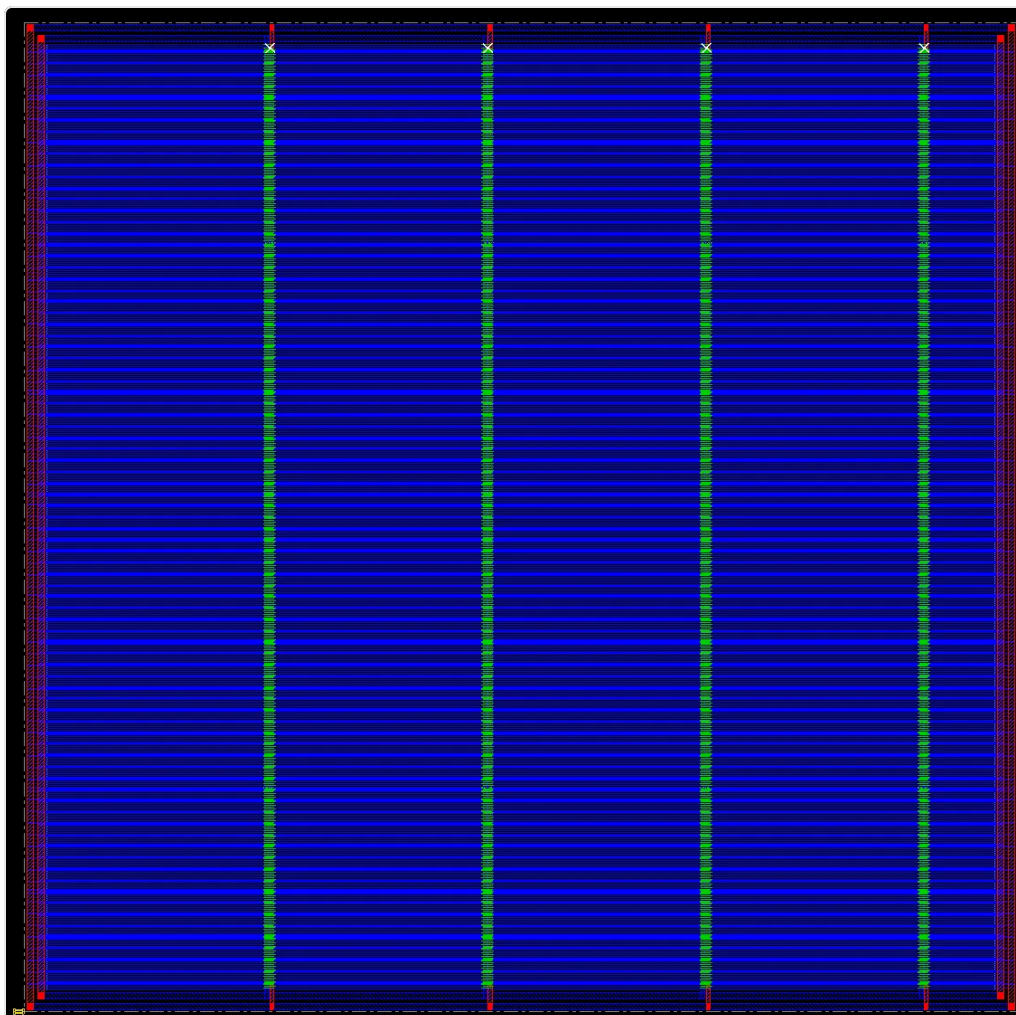
☐ Check Standard Cell Geometry

☐ Split vias longer than 0 into smaller vias
with center-to-center step of 0 and height -1
and bottom/left edge offset of -1

Make Via Connections To:

☒ Pad Ring/Pin ☒ Stripe ☒ Block Pin ☐ Core Wire ☐ IO Wire
☒ Core Ring ☒ Block Ring ☐ Block Wire ☐ Follow Pin ☐ No Shape

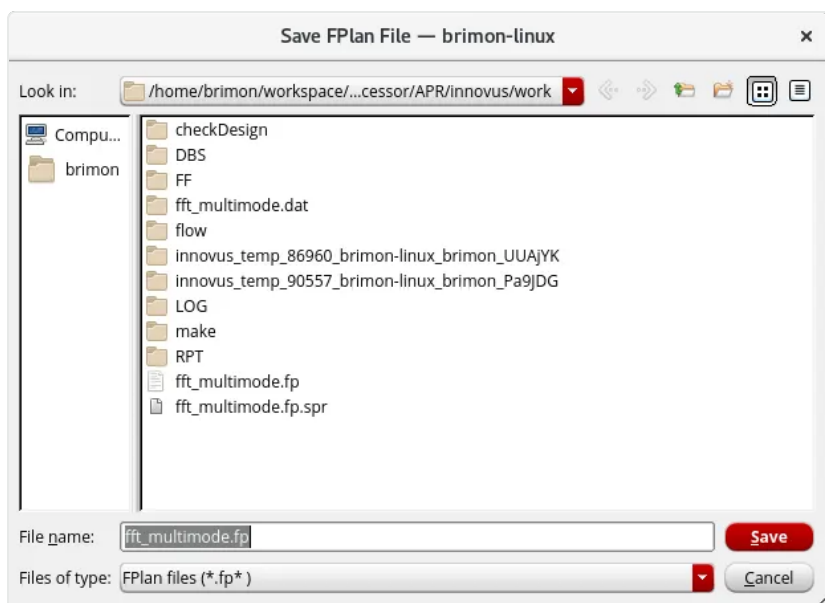
OK Apply Defaults Cancel Help



floorplan

随后还需要对 Pin 进行编辑，这里也就先跳过了。

结束上面的步骤并且没有报错后，需要保存 floorplan 用于后续的操作：



PnR

Floorplan 完成后进行后续的操作：

打开 innovus， Flow -> **Create Foundation Flow Template** -> save 保存模板到 work/flow 文件夹。

([无法找到Innovus Flow Wizard](#)： innovus21 及以上的版本没有 wizard， 只能生成模板并用 setup.tcl 配置)

([Innovus 中 Foundation Flow简介](#))

创建 setup.tcl 配置文件：

```

1  set design_name fft_multimode
2  set vars(design) $design_name
3  set vars(rtl_design) $vars(design)
4  set init_verilog "../fft_multimode.v"
5  set vars(fp_file) "fft_multimode.fp"
6
7  set init_lef_file "/tools/PDK/tsmc28nm/tn28clpr002e1_1_5a/N28_PRTF_Cad_v1d
5a/PR_tech/Cadence/LefHeader/HVH/tsmcn28_10lm7X2ZRDL.tlef"
8  set vars(lef_files) "/tools/PDK/tsmc28nm/tn28clpr002e1_1_5a/N28_PRTF_Cad_v
1d5a/PR_tech/Cadence/LefHeader/HVH/tsmcn28_10lm7X2ZRDL.tlef \
9    /tools/PDK/tsmc28nm/TSMCHOME/digital/Back_End/lef/tcbn28hpcplusbwp40p14
0_110a/lef/tcbn28hpcplusbwp40p140.lef"
10 set init_top_cell $vars(design)
11 set init_gnd_net VSS
12 set init_pwr_net VDD
13 set vars(cts_cells) clk
14
15 set vars(local_cpus) 8
16 set vars(process) 28;# e.g. 16, 28
17
18 set vars(rc_corners) "cworst_m40c cbest_125c ctyp_85c cworst_125c"
19 set vars(delay_corners) "dc_slow_corner_cworst_m40c dc_typ_corner_ctyp_85
c dc_fast_corner_cbest_125c"
20 set vars(constraint_modes) "merged" ;# set vars(constraint_modes) "merged
merged_1g"
21
22 set slow_corner      ssgnp_0p90v_m40c ;# e.g. ssgnp_0p72v_0c
23 set typ_corner       tt_1p00v_85c      ;# e.g. tt_0p80v_85c
24 set fast_corner      ffgnp_1p05v_125c ;# e.g. ffgnp_0p88v_125c
25
26 ▼ set vars(merged_slow_corner_cworst_m40c,delay_corner) [lindex $vars(de
lay_corners) 0]
27 ▼ set vars(merged_fast_corner_cbest_125c,delay_corner) [lindex $vars(de
lay_corners) 2]
28 ▼ set vars(merged_slow_corner_cworst_m40c,constraint_mode) [lindex $vars(co
nstraint_modes) 0]
29 ▼ set vars(merged_fast_corner_cbest_125c,constraint_mode) [lindex $vars(co
nstraint_modes) 0]
30 ▼ set vars(dc_fast_corner_cbest_125c,rc_corner) [lindex $vars(rc
_corners) 1]
31 ▼ set vars(dc_typ_corner_ctyp_85c,rc_corner) [lindex $vars(rc
_corners) 2]
32 set vars(dc_fast_corner_cbest_125c,library_set) $fast_corner
33 set vars(dc_typ_corner_ctyp_85c,library_set) $typ_corner
34 set vars(dc_slow_corner_cworst_m40c,library_set) $slow_corner

```

```

35  set vars(dc_slow_corner_cworst_m40c,rc_corner) [lindex $vars(rc
36  _corners) 0]
37
38  set vars(setup_analysis_views) "merged_slow_corner_cworst_m40c"
39  set vars(hold_analysis_views)  "merged_fast_corner_cbest_125c"
40  set vars(setup_analysis_views) "merged_slow_corner_cworst_m40c"
41  set vars(hold_analysis_views)  "merged_fast_corner_cbest_125c"
42
43  set vars(default_setup_view)    [lindex $vars(setup_analysis_views) 0]
44  set vars(default_hold_view)     [lindex $vars(hold_analysis_views) 0]
45  set vars(active_setup_views)    $vars(setup_analysis_views)
46  set vars(active_hold_views)     $vars(hold_analysis_views)
47
48  set qrc_file(cworst)  "/tools/PDK/tsmc28nm/RC_QRC_cln28hpc+_1p10m_5x2y2z_
49  ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10m+ut-alrdl_5x2y2z_cworst_T/qrc
50  TechFile"
51  set qrc_file(cbest)   "/tools/PDK/tsmc28nm/RC_QRC_cln28hpc+_1p10m_5x2y2z_
52  ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10m+ut-alrdl_5x2y2z_cbest/qrcTec
53  hFile"
54  set qrc_file(rcworst) "/tools/PDK/tsmc28nm/RC_QRC_cln28hpc+_1p10m_5x2y2z_
55  ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10m+ut-alrdl_5x2y2z_rcworst_T/qr
56  cTechFile"
57  set qrc_file(rcbest)  "/tools/PDK/tsmc28nm/RC_QRC_cln28hpc+_1p10m_5x2y2z_
58  ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10m+ut-alrdl_5x2y2z_rcbest/qrcTe
59  chFile"
60  set qrc_file(typical) "/tools/PDK/tsmc28nm/RC_QRC_cln28hpc+_1p10m_5x2y2z_
61  ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10m+ut-alrdl_5x2y2z_typical/qrcT
62  echFile"
63  set vars(cworst_m40c,qx_tech_file) $qrc_file(cworst)
64  set vars(cbest_125c,qx_tech_file)  $qrc_file(cbest)
65  set vars(ctyp_85c,qx_tech_file)    $qrc_file(typical)
66  set vars(cworst_125c,qx_tech_file) $qrc_file(cworst)
67
68  set stdcell_library($fast_corner)  "/tools/PDK/tsmc28nm/TSMCHOME/digital/
69  Front_End/timing_power_noise/NLDM/tcbn28hpcplusbwp40p140_180a/tcbn28hpcplu
70  sbwp40p140ffg0p88v0c.lib"
71  set stdcell_library($slow_corner)  "/tools/PDK/tsmc28nm/TSMCHOME/digital/
72  Front_End/timing_power_noise/NLDM/tcbn28hpcplusbwp40p140_180a/tcbn28hpcplu
73  sbwp40p140ffg0p88v0c.lib"
74  set stdcell_library($typ_corner)    "/tools/PDK/tsmc28nm/TSMCHOME/digital/
75  Front_End/timing_power_noise/NLDM/tcbn28hpcplusbwp40p140_180a/tcbn28hpcplu
76  sbwp40p140ffg0p88v0c.lib"
77
78  set macro_ctl_list [list ]
79  set macro(lef)     [list ]
80  set macro_library($slow_corner) [list ]
81  set macro_library($typ_corner)  [list ]
82  set macro_library($fast_corner) [list ]

```

```

66 set vars(library_sets) [list $slow_corner $typ_corner $fast_corner]
67 set vars($slow_corner,timing) [concat $stdcell_library($slow_corner) $macro_library($slow_corner)]
68 set vars($typ_corner,timing) [concat $stdcell_library($typ_corner) $macro_library($typ_corner)]
69 set vars($fast_corner,timing) [concat $stdcell_library($fast_corner) $macro_library($fast_corner)]
70
71 set vars(merged,pre_cts_sdc) "../fft_multimode.sdc"
72

```

生成脚本：

```

▼ Foundation Flow Shell |
1 tclsh ./flow/SCRIPTS/gen_flow.tcl -m flat all

```

会在 work 文件夹中生成 makefile 以及 FF 文件夹等文件。

生成完脚本和 makefile 后就可以进行后面的流程了：

```

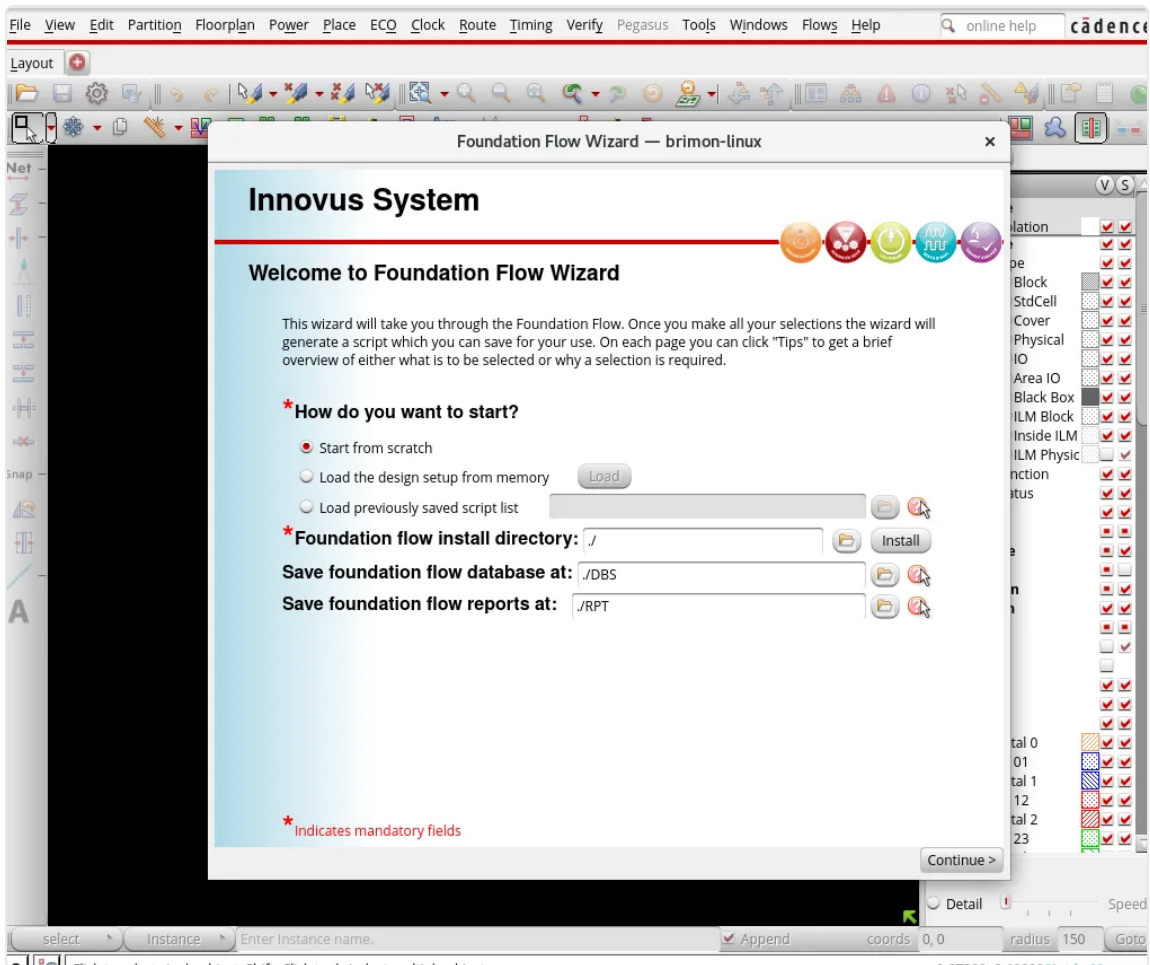
▼ Shell |
1 make help # 查看make的选项进行对应的操作
2
3 make init # 导入设计并进行初始化
4 make place
5 make cts
6 make postcts_hold # CTS后修复hold
7 make route
8 make signoff

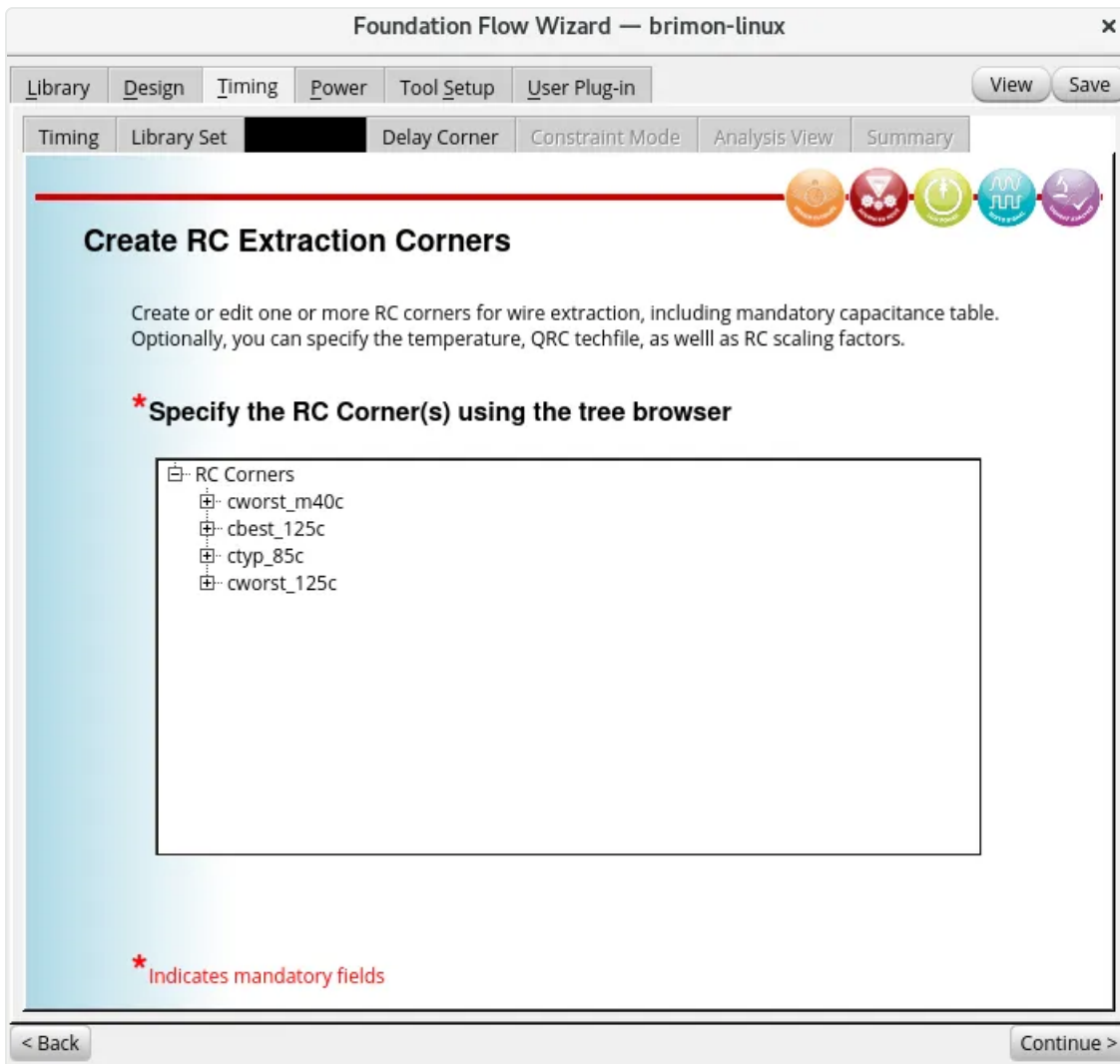
```

另外：在 make init 的过程中可能会报错 cell 错误，这时可以修改 FF/init.tcl 中的 init_top_cell 并在执行 make 时加上 UPDATE=no，不知道为什么脚本在设置的时候会加上地址。

Foundation Flow Wizard

20 及以下的版本的 innovus 有 **Foundation Flow Wizard** 可以辅助生成 setup.tcl 的配置，跟着引导走进行选项的设置，这个有助于理解 setup.tcl 中每个参数是什么意思。






```

1  global vars
2  set vars(version)                20.1.0
3
4  #####
5  # Define some variables to point to data, libraries, and scripts
6  #-----
7  #set vars(design_root)            ""
8  set vars(script_root)            ./SCRIPTS
9  set vars(dbs_dir)                ./DBS
10 set vars(rpt_dir)                 ./RPT
11
12
13 #####
14 # Define technology and physical libraries
15 #-----
16 set vars(process)                 28
17 set vars(lef_files) \
18   "/tools/PDK/tsmc28nm/tn28clpr002e1_1_5a/N28_PRTF_Cad_v1d5a/PR_tech/Cade
   nce/LefHeader/HVH/tsmcn28_10lm7X2ZRDL.tlef \
19   /tools/PDK/tsmc28nm/TSMCHOME/digital/Back_End/lef/tcbn28hpcplusbwp40p1
   40_110a/lef/tcbn28hpcplusbwp40p140.lef \
20   /tools/PDK/tsmc28nm/TSMCHOME/sram/Compiler/tsn28hpcpd127spsram_2012020
   0_180a/sramsp16x256_tsmc28hpc/LEF/sramsp16x256_tsmc28hpc.lef"
21 #set vars(ilm_list)               ""
22 #set vars(partition_list)         ""
23
24
25 #####
26 # Define the design data
27 #-----
28 set vars(netlist) "../fft_multimode.v"
29 set vars(design)                  fft_multimode
30 set vars(fp_file) "fft_multimode.fp"
31 #set vars(def_files) ""
32 #set vars(fp_tcl_file) ""
33 #set vars(fp_tcl_proc)            ""
34 #set vars(scan_def) ""
35 #set vars(power_nets)             ""
36 #set vars(ground_nets)           ""
37 set vars(honor_pitch)             FALSE
38
39 #set vars(cts_spec) ""
40 set vars(cts_cells)               clk
41
42 set vars(enable_ss)               false

```

```

43 set vars(flow)                                mmmc
44 set vars(enable_ocv)                          pre_postcts
45
46
47 #####
48 # Define library sets ...
49 #-----
50 set vars(library_sets)                        "ssgnp_0p90v_m40c tt_1
51 p00v_85c ffgnp_1p05v_125c"
52 #set vars(ssgnp_0p90v_m40c,si) ""
53 set vars(ssgnp_0p90v_m40c,timing) "/tools/PDK/tsmc28nm/TSMCHOME/digital/F
54 ront_End/timing_power_noise/NLDM/tcbn28hpcplusbwp40p140_180a/tcbn28hpcplu
55 sbwp40p140ffg0p88v0c.lib"
56 #set vars(tt_1p00v_85c,si) ""
57 set vars(tt_1p00v_85c,timing) "/tools/PDK/tsmc28nm/TSMCHOME/digital/Front
58 _End/timing_power_noise/NLDM/tcbn28hpcplusbwp40p140_180a/tcbn28hpcplusbwp
59 40p140ffg0p88v0c.lib"
60 #set vars(ffgnp_1p05v_125c,si) ""
61 set vars(ffgnp_1p05v_125c,timing) "/tools/PDK/tsmc28nm/TSMCHOME/digital/F
62 ront_End/timing_power_noise/NLDM/tcbn28hpcplusbwp40p140_180a/tcbn28hpcplu
63 sbwp40p140ffg0p88v0c.lib"
64
65 #####
66 # Define rc corners ...
67 #-----
68 set vars(rc_corners)                          "cworst_m40c cbest_12
69 5c ctyp_85c cworst_125c"
70 #set vars(cworst_m40c,qx_lib_file)            ""
71 #set vars(cworst_m40c,qx_conf_file)           ""
72 #set vars(cworst_m40c,cap_table)              ""
73 set vars(cworst_m40c,qx_tech_file)            /tools/PDK/tsmc28nm/R
74 C_QRC_cln28hpc+_1p10m_5x2y2z_ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10
75 m+ut-alrdl_5x2y2z_cworst_T/qrcTechFile
76 #set vars(cworst_m40c,T)                      ""
77 #set vars(cworst_m40c,pre_route_res_factor)   ""
78 #set vars(cworst_m40c,pre_route_cap_factor)   ""
79 #set vars(cworst_m40c,pre_route_clk_res_factor) ""
80 #set vars(cworst_m40c,pre_route_clk_cap_factor) ""
81 #set vars(cworst_m40c,post_route_res_factor)  ""
82 #set vars(cworst_m40c,post_route_cap_factor)  ""
83 #set vars(cworst_m40c,post_route_xcap_factor) ""
84 #set vars(cworst_m40c,post_route_clk_res_factor) ""
85 #set vars(cworst_m40c,post_route_clk_cap_factor) ""
86 #set vars(cworst_m40c,xcap_factor)            ""
87 #set vars(cbest_125c,qx_lib_file)            ""
88 #set vars(cbest_125c,qx_conf_file)           ""
89 #set vars(cbest_125c,cap_table)              ""

```

```

81  set vars(cbest_125c,qx_tech_file) /tools/PDK/tsmc28nm/R
C_QRC_cln28hpc+_1p10m_5x2y2z_ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10
m+ut-alrdl_5x2y2z_cbest/qrcTechFile
82  #set vars(cbest_125c,T) ""
83  #set vars(cbest_125c,pre_route_res_factor) ""
84  #set vars(cbest_125c,pre_route_cap_factor) ""
85  #set vars(cbest_125c,pre_route_clk_res_factor) ""
86  #set vars(cbest_125c,pre_route_clk_cap_factor) ""
87  #set vars(cbest_125c,post_route_res_factor) ""
88  #set vars(cbest_125c,post_route_cap_factor) ""
89  #set vars(cbest_125c,post_route_xcap_factor) ""
90  #set vars(cbest_125c,post_route_clk_res_factor) ""
91  #set vars(cbest_125c,post_route_clk_cap_factor) ""
92  #set vars(cbest_125c,xcap_factor) ""
93  #set vars(ctyp_85c,qx_lib_file) ""
94  #set vars(ctyp_85c,qx_conf_file) ""
95  #set vars(ctyp_85c,cap_table) ""
96  set vars(ctyp_85c,qx_tech_file) /tools/PDK/tsmc28nm/R
C_QRC_cln28hpc+_1p10m_5x2y2z_ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10
m+ut-alrdl_5x2y2z_typical/qrcTechFile
97  #set vars(ctyp_85c,T) ""
98  #set vars(ctyp_85c,pre_route_res_factor) ""
99  #set vars(ctyp_85c,pre_route_cap_factor) ""
100 #set vars(ctyp_85c,pre_route_clk_res_factor) ""
101 #set vars(ctyp_85c,pre_route_clk_cap_factor) ""
102 #set vars(ctyp_85c,post_route_res_factor) ""
103 #set vars(ctyp_85c,post_route_cap_factor) ""
104 #set vars(ctyp_85c,post_route_xcap_factor) ""
105 #set vars(ctyp_85c,post_route_clk_res_factor) ""
106 #set vars(ctyp_85c,post_route_clk_cap_factor) ""
107 #set vars(ctyp_85c,xcap_factor) ""
108 #set vars(cworst_125c,qx_lib_file) ""
109 #set vars(cworst_125c,qx_conf_file) ""
110 #set vars(cworst_125c,cap_table) ""
111 set vars(cworst_125c,qx_tech_file) /tools/PDK/tsmc28nm/R
C_QRC_cln28hpc+_1p10m_5x2y2z_ut-alrdl_9corners_1.3a/RC_QRC_cln28hpc+_1p10
m+ut-alrdl_5x2y2z_cworst_T/qrcTechFile
112 #set vars(cworst_125c,T) ""
113 #set vars(cworst_125c,pre_route_res_factor) ""
114 #set vars(cworst_125c,pre_route_cap_factor) ""
115 #set vars(cworst_125c,pre_route_clk_res_factor) ""
116 #set vars(cworst_125c,pre_route_clk_cap_factor) ""
117 #set vars(cworst_125c,post_route_res_factor) ""
118 #set vars(cworst_125c,post_route_cap_factor) ""
119 #set vars(cworst_125c,post_route_xcap_factor) ""
120 #set vars(cworst_125c,post_route_clk_res_factor) ""
121 #set vars(cworst_125c,post_route_clk_cap_factor) ""
122 #set vars(cworst_125c,xcap_factor) ""

```

```

123
124
125 #####
126 # Define delay corners ...
127 #-----
128 set vars(enable_cpvr) none
129 set vars(delay_corners) "dc_slow_corner_
130 cworst_m40c dc_typ_corner_ctyp_85c dc_fast_corner_cbest_125c"
131 #set vars(dc_slow_corner_cworst_m40c,clock_cell_early) ""
132 #set vars(dc_slow_corner_cworst_m40c,clock_cell_late) ""
133 #set vars(dc_slow_corner_cworst_m40c,clock_net_early) ""
134 #set vars(dc_slow_corner_cworst_m40c,clock_net_late) ""
135 #set vars(dc_slow_corner_cworst_m40c,data_cell_early) ""
136 #set vars(dc_slow_corner_cworst_m40c,data_cell_late) ""
137 #set vars(dc_slow_corner_cworst_m40c,data_net_early) ""
138 #set vars(dc_slow_corner_cworst_m40c,data_net_late) ""
139 set vars(dc_slow_corner_cworst_m40c,rc_corner) cworst_m40c
140 set vars(dc_slow_corner_cworst_m40c,library_set) ssgnp_0p90v_m40c
141 set vars(dc_slow_corner_cworst_m40c,cell_check_late) ssgnp_0p90v_m40c
142 set vars(dc_slow_corner_cworst_m40c,cell_check_early) ssgnp_0p90v_m40c
143 #set vars(dc_typ_corner_ctyp_85c,clock_cell_early) ""
144 #set vars(dc_typ_corner_ctyp_85c,clock_cell_late) ""
145 #set vars(dc_typ_corner_ctyp_85c,clock_net_early) ""
146 #set vars(dc_typ_corner_ctyp_85c,clock_net_late) ""
147 #set vars(dc_typ_corner_ctyp_85c,data_cell_early) ""
148 #set vars(dc_typ_corner_ctyp_85c,data_cell_late) ""
149 #set vars(dc_typ_corner_ctyp_85c,data_net_early) ""
150 #set vars(dc_typ_corner_ctyp_85c,data_net_late) ""
151 set vars(dc_typ_corner_ctyp_85c,rc_corner) ctyp_85c
152 set vars(dc_typ_corner_ctyp_85c,library_set) tt_1p00v_85c
153 set vars(dc_typ_corner_ctyp_85c,cell_check_late) tt_1p00v_85c
154 set vars(dc_typ_corner_ctyp_85c,cell_check_early) tt_1p00v_85c
155 #set vars(dc_fast_corner_cbest_125c,clock_cell_early) ""
156 #set vars(dc_fast_corner_cbest_125c,clock_cell_late) ""
157 #set vars(dc_fast_corner_cbest_125c,clock_net_early) ""
158 #set vars(dc_fast_corner_cbest_125c,clock_net_late) ""
159 #set vars(dc_fast_corner_cbest_125c,data_cell_early) ""
160 #set vars(dc_fast_corner_cbest_125c,data_cell_late) ""
161 #set vars(dc_fast_corner_cbest_125c,data_net_early) ""
162 #set vars(dc_fast_corner_cbest_125c,data_net_late) ""
163 set vars(dc_fast_corner_cbest_125c,rc_corner) cbest_125c
164 set vars(dc_fast_corner_cbest_125c,library_set) ffgnp_1p05v_125c
165 set vars(dc_fast_corner_cbest_125c,cell_check_late) ffgnp_1p05v_125c
166 set vars(dc_fast_corner_cbest_125c,cell_check_early) ffgnp_1p05v_125c
167
168 #####
169 # Define constraints modes ...

```

```

170 #-----
171 set vars(constraint_modes) merged
172 set vars(merged,pre_cts_sdc) "../fft_multimode.sdc"
173 #set vars(merged,incr_cts_sdc) ""
174 #set vars(merged,post_cts_sdc) ""
175 #set vars(merged,ilm_pre_cts_sdc) ""
176 #set vars(merged,ilm_incr_cts_sdc) ""
177 #set vars(merged,ilm_post_cts_sdc) ""
178
179
180 #####
181 # Define analysis views ...
182 #-----
183 set vars(default_hold_view) merged_fast_
184   corner_cbest_125c
185 set vars(default_setup_view) merged_slow_
186   corner_cworst_m40c
187 set vars(hold_analysis_views) merged_fast_
188   corner_cbest_125c
189   set vars(merged_fast_corner_cbest_125c,constraint_mode) merged
190   set vars(merged_fast_corner_cbest_125c,delay_corner) dc_fast_corn
191   er_cbest_125c
192   set vars(setup_analysis_views) merged_slow_
193   corner_cworst_m40c
194   set vars(merged_slow_corner_cworst_m40c,constraint_mode) merged
195   set vars(merged_slow_corner_cworst_m40c,delay_corner) dc_slow_corn
196   er_cworst_m40c
197   set vars(active_hold_views) merged_fast_
198   corner_cbest_125c
199   set vars(active_setup_views) merged_slow_
200   corner_cworst_m40c
201
202 #####
203 # Define power settings ...
204 #-----
205 set vars(power_analysis_view) merged_slow_corner_cwo
206   rst_m40c
207 set vars(leakage_power_effort) none
208 set vars(dynamic_power_effort) none
209 #set vars(activity_file) ""
210 set vars(activity_file_format) TCF
211 set vars(report_power) FALSE
212 #set vars(cpf_file) ""
213 set vars(cpf_keep_rows) FALSE
214 set vars(cpf_power_domain) FALSE
215 set vars(cpf_power_switch) FALSE
216 set vars(cpf_isolation) FALSE

```

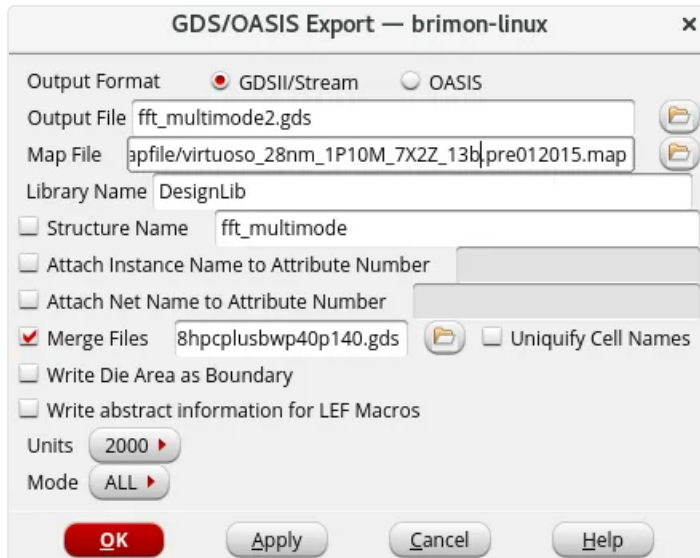
```

209 set vars(cpf_state_retention) FALSE
210 set vars(cpf_level_shifter) FALSE
211
212
213 #####
214 # Define tool specific options ...
215 #-----
216
217 set vars(max_route_layer) 11
218 set vars(generate_tracks) TRUE
219 set vars(postroute_extraction_effort) low
220 set vars(multi_cut_effort) low
221 set vars(litho_driven_routing) FALSE
222 set vars(postroute_spread_wires) FALSE
223
224 #set vars(delta_delay_threshold) ""
225 #set vars(celtic_settings) ""
226 #set vars(coupling_c_thresh) ""
227 #set vars(relative_c_thresh) ""
228 #set vars(total_c_thresh) ""
229 set vars(si_analysis_type) default
230 set vars(signoff_extraction_effort) high
231 #set vars(antenna_diode) ""
232
233 set vars(metalfill) false
234 #set vars(metalfill_tcl) ""
235 #set vars(gds_files) ""
236 #set vars(qrc_layer_map) ""
237 #set vars(qrc_library) ""
238 #set vars(qrc_config_file) ""
239 #set vars(gds_layer_map) ""
240 #set vars(oa_abstract_name) ""
241 #set vars(oa_layout_name) ""
242 #set vars(oa_ref_lib) ""
243 #set vars(oa_design_lib) ""
244 #set vars(oa_design_cell) ""
245 #set vars(oa_design_view) ""
246
247 set vars(local_cpus) 8
248 #set vars(remote_hosts) ""
249 #set vars(cpu_per_remote_host) ""
250 set vars(distribute) local
251 #set vars(custom_script) ""
252 #set vars(lsf_queue) ""
253 #set vars(lsf_resource) ""
254 #set vars(lsf_args) ""
255 #set vars(rsh_host_list) ""
256 #set vars(distribute_timeout) ""

```

生成 setup.tcl 后按照前面的步骤执行即可。

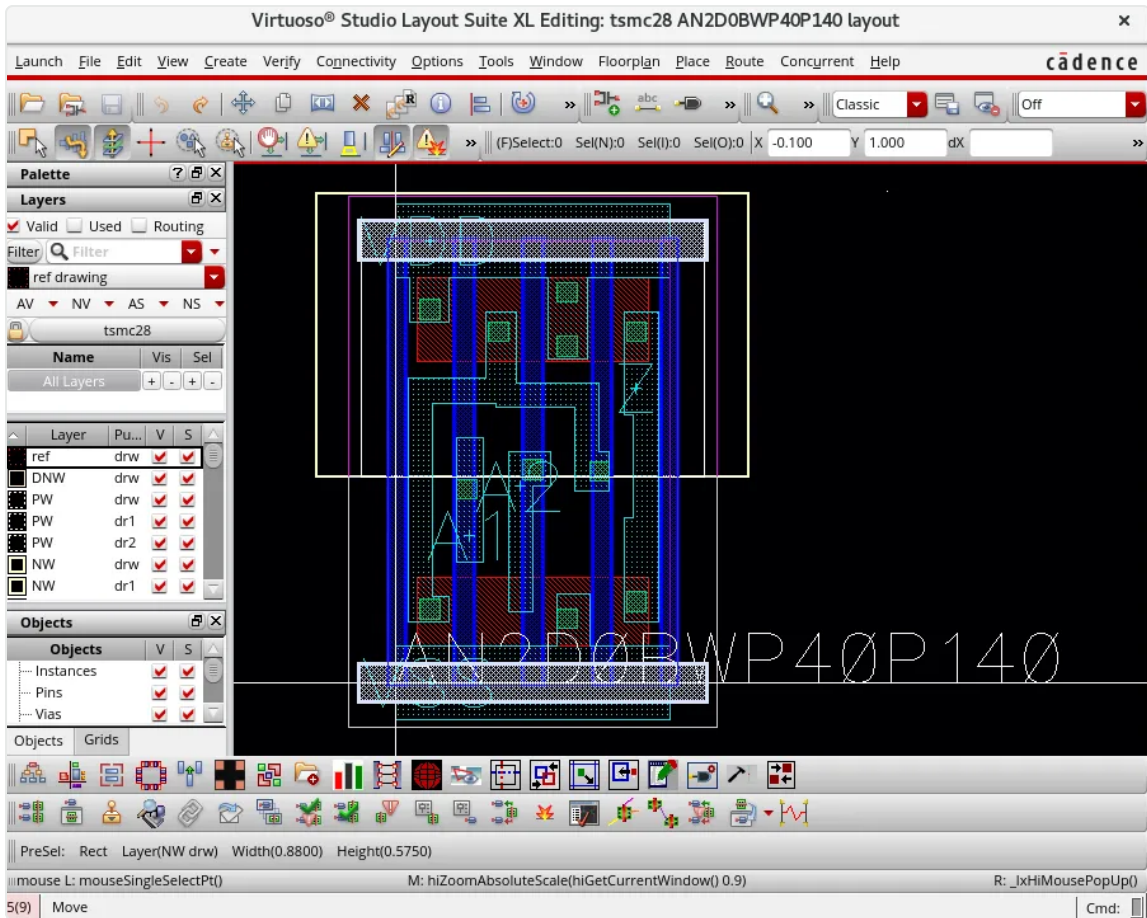
导出 GDS: File -> Save



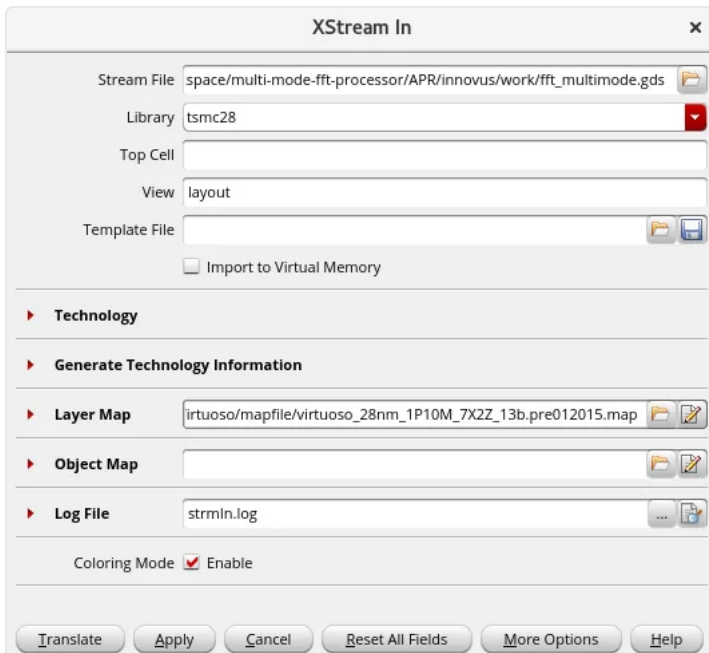
Virtuoso 导入 library: 在 Tools -> Library Path Editor 中加入路径:

*/TSMCHOME/digital/Back_End/cdk/tcbn28hpcplusbwp40p140_110a/tcbn28hpcplusbwp40p140

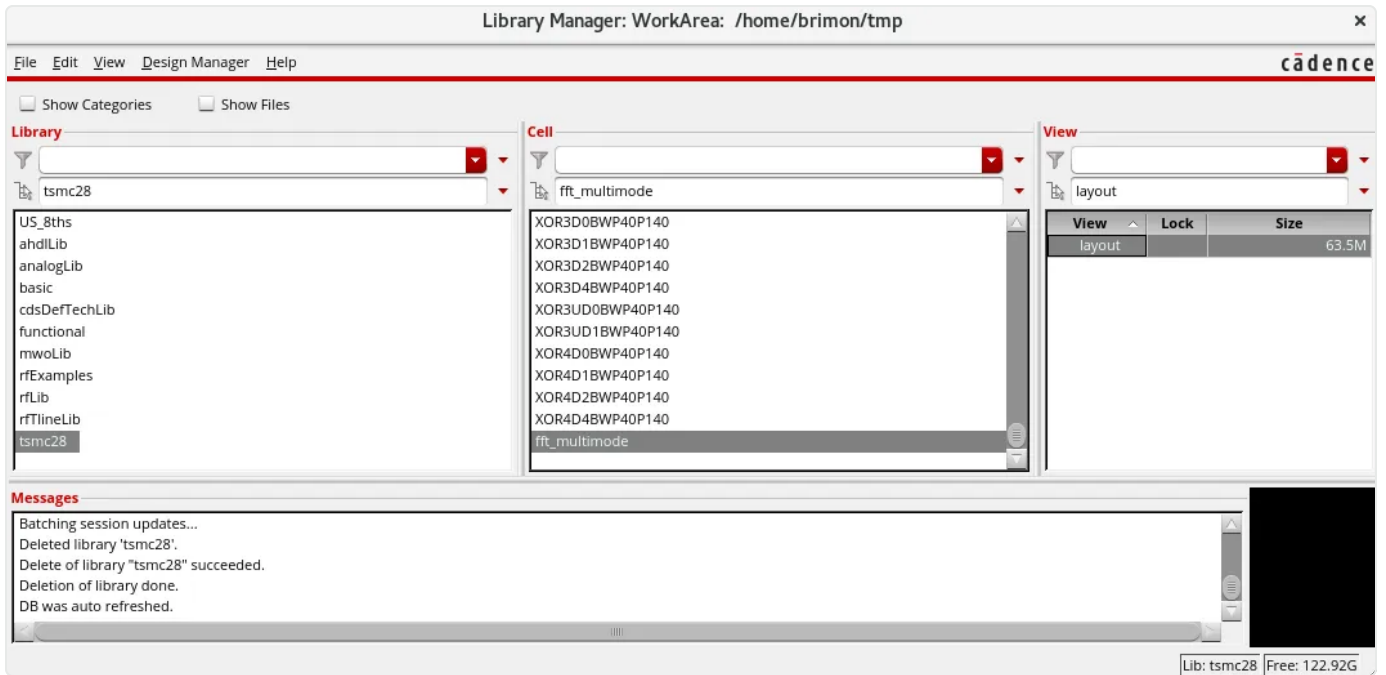
在 Tools -> Display Resource Manager 中载入 display.drf 文件, 然后就可以正常显示 layout, 例如一个 AN2D 的 layout 如下:



然后 File -> import -> stream 选择导出的 gds 文件和刚才创建的 tsmc28 的 library。
这里的 Layer map 要选择和导出的时候一样的 map 文件

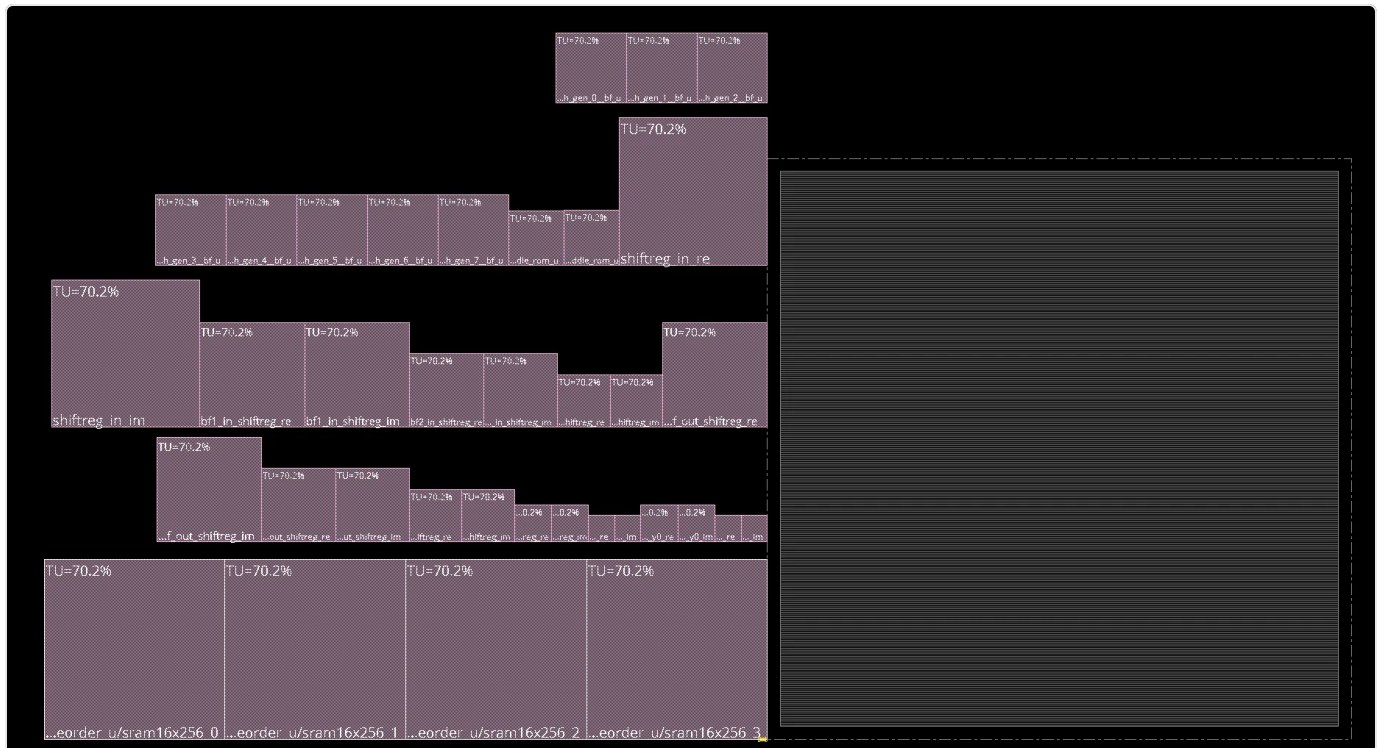


打开导入的 gds 的 layout:

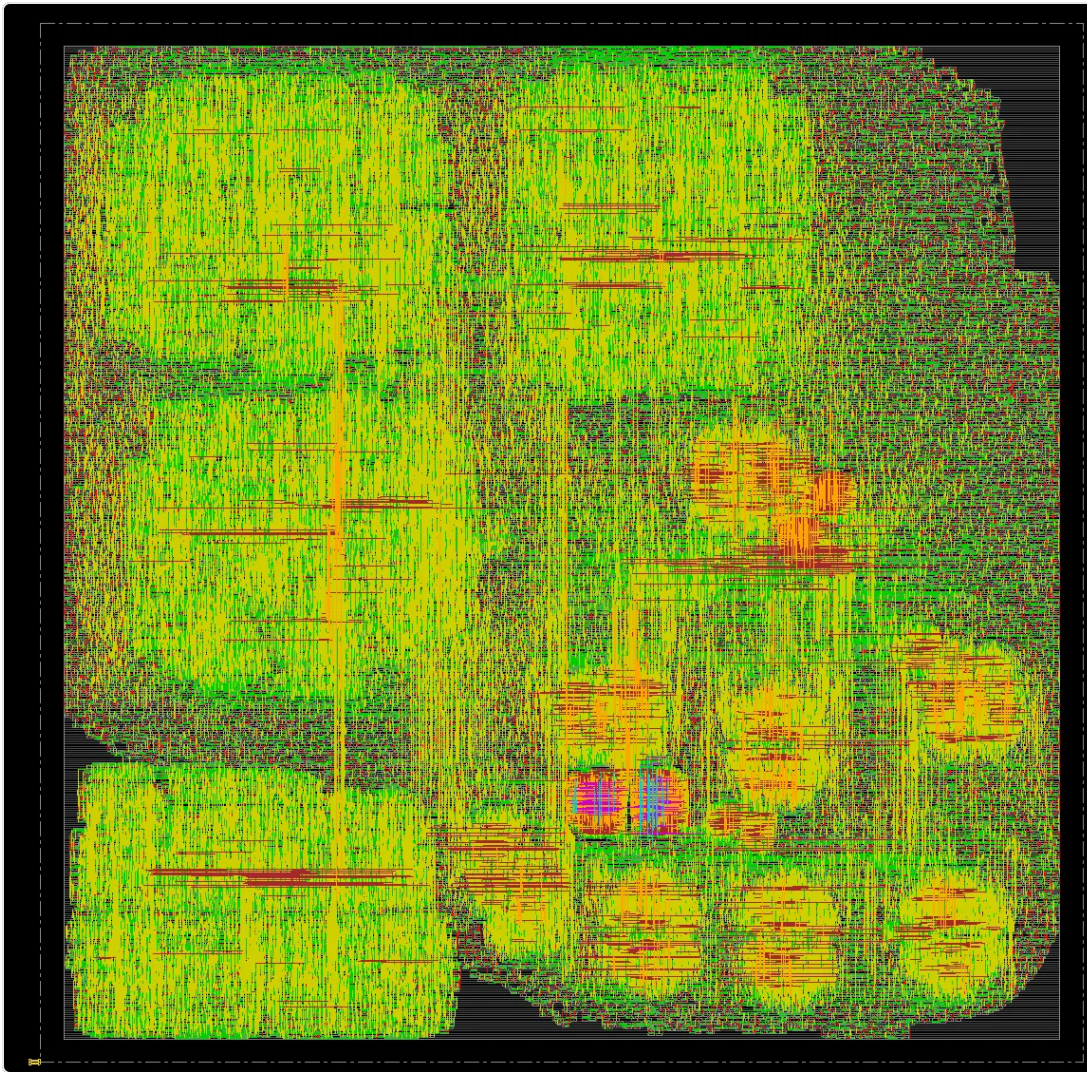


结果

可以通过 File -> Restore Design 读取存在 work/DBS 中的结果。



Place



Route

